

Building and Calibrating a Hybrid RAG Pipeline

Technical Report on Indexed Retrieval Augmented Generation

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 [hybrid-rag-playground](https://github.com/hybrid-rag-playground)

1 Pipeline Overview

Here, this report presents a retrieval augmented generation (RAG) pipeline end-to-end, across three BEIR datasets (NFCorpus, SciFact, and FiQA) which integrates sparse, dense, and hybrid search methods along with a small LLM, running locally. Everything is offline here: **Weaviate** (Docker) stores chunk databases in BYO-vector format, **sentence-transformers/all-MiniLM-L6-v2** creates 384-dimensional chunks, deterministically caching the result on disk and then inserting into Weaviate database, and **Ollama** hosts the generator itself (default: Gemma 2 2B; optional: Llama 3.2 3B / Llama 3.1 8B). The FastAPI backend runs all these services and the React + TypeScript frontend lets you tweak every parameter manually.

Documents are split via *adaptive document-aware chunker*: short documents are left intact; longer ones have sentences taken with up to 512-tokens sliding window, overlapping by around 10%. When a single sentence is longer than this budget, it commutes back to token window splitting instead. Every chunk gets embedded once, cached by its hash and added to respective dataset Weaviate database. Automatic indexing of BM25 over chunk texts takes place here as well. See below Figure 1.

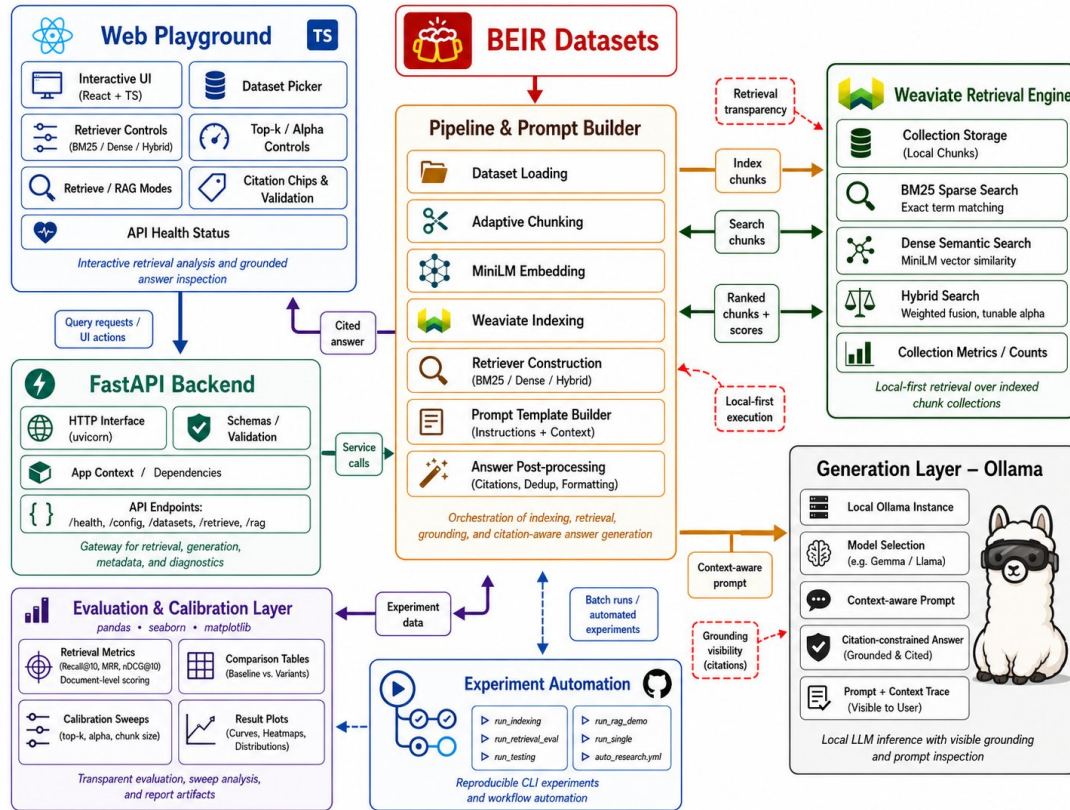


Figure 1: End-to-end pipeline: BEIR dataset goes into the adaptive chunker; chunks are embodied using MiniLM and indexed with Weaviate (BM25, dense, and hybrid approach). The FastAPI serves `/retrieve` and `/rag` endpoints; Ollama gives citation-supported responses; the playground analyzes everything for each stage.

In general, the architecture has been designed not only as a retrieval benchmarking mechanism but also as a diagnostic one in order to identify cases where RAG succeeds and where it falls short. Through its use of clear delineations between indexing, retrieval, ranking, prompt formation, generation, and citation validation steps, the pipeline can track down the provenance of each generated response all the way down to the chunks used to retrieve it. The reason for doing so is that the quality of RAG responses is contingent upon not only the generator but also chunking, embeddings, lexical matching, hybrid fusion, and prompting.

Hybrid fusion

As opposed to combining incompatible BM25 and cosine similarity scores, the hybrid retriever combines *Reciprocal Rank Fusion* (*RRF*) with weights of:

$$\text{score}(d) = \frac{1 - \alpha}{k + r_{\text{bm25}}(d)} + \frac{\alpha}{k + r_{\text{dense}}(d)}, \quad k = 60.$$

Each retriever retrieves $5 \times \text{top-}k$, the merged list is cut down to $\text{top-}k$. It is scale independent and evaluated using BEIR qrels at the *document* level (a document is deemed to be retrieved if any of its chunks makes it to $\text{top-}k$ value selected by the user).

2 Retriever Comparison

For each dataset, it is performed a complete test run through the three information retrieval systems using the suggested $\alpha=0.5$ and $\text{top-}k=10$. The evaluation metrics such as Recall@10, MMR, and nDCG@10 were calculated based on a 192-token chunk budget (Table 1).

Retriever	NFCorpus (323 q)			SciFact (300 q)			FiQA (648 q)		
	R@10	MRR	nDCG@10	R@10	MRR	nDCG@10	R@10	MRR	nDCG@10
BM25	0.125	0.481	0.254	0.706	0.579	0.602	0.206	0.205	0.161
Dense	0.134	0.488	0.279	0.785	0.621	0.656	0.442	0.452	0.375
Hybrid	0.145	0.536	0.300	0.773	0.632	0.662	0.385	0.371	0.305

Table 1: Retriever comparison across the three BEIR datasets ($\text{top-}k=10$, $\alpha=0.5$, 192-token chunks).

Discussion

The lack of an unequivocal best retriever on all 3 datasets is consistent with BEIR findings.

Three regimes are revealed:

- **SciFact (scientific fact-checking)**: claims and abstracts are from the same language space ("vitamin B12", "perinatal mortality"), hence BM25 starts off with decent performance ($\text{nDCG@10}=0.602$). Dense provides paraphrasing capability, while hybrid provides optimal balance of the two approaches ($\text{nDCG@10}=\mathbf{0.662}$; $\text{MRR}=\mathbf{0.632}$). The fusion wins due to the complementarity of both metrics results in short, proper claims.
- **NFCorpus (medical / nutrition)**: short queries, many of which are information needs stated naturally ("vitamin D and prostate cancer"). By itself, BM25 struggles ($\text{R@10}=0.125$), since the query-abstract vocabulary mismatch is high. Hybrid fusion wins on *all three* metrics ($\text{R@10}+\mathbf{16\%}$ vs. BM25, $\text{MRR}+\mathbf{11\%}$ vs. dense), showing that BM25 still contributes precision in the rare matching cases.
- **FiQA (financial Q&A)**: conversational, noisy forum Q's on a corpus with only 57k documents and poor overlap of terms with answers. *Density dominates by a wide margin* ($\text{nDCG@10}=0.375$ vs. 0.161 nDCG@10 , $+133\%$ improvement over BM25). Hybrid ($\alpha=0.5$) is severely affected by poor performance of the BM25 list. As seen from sweep experiments, increasing α to 1.0 reduces this gap considerably, which means that indicating that the *default* $\alpha=0.5$ is the wrong operating point here, not that hybrid fusion is intrinsically bad.

Takeaway. Hybrid retrieval is a strong default but not a universal winner; the right α is corpus-dependent and must be calibrated rather than fixed at 0.5.

3 End-to-End RAG on SciFact

SciFact was chosen for use in the end-to-end RAG demo: hybrid retrieval performs better than others in two out of three metrics, claims are nicely formulated and make good sense for citation answers, and the corpus size ($\sim 5\text{k}$ abstracts) allows efficient runs on laptops. For retriever, it is using hybrid method with $\alpha=0.5$ and $\text{top-}k=5$; for generator, Gemma 2 2B by Ollama. The prompt consisted of the user claim, five retrieved chunks with `chunk_id` tags, and an instruction to cite the IDs as-is; IDs not matched were detected during citations validation as *invented*. Manually assessed 10 test claims; see Table 2 for 5 examples.

#	Claim	Top chunks (id)	Verdict
1	0-dim biomaterials show inductive properties.	803312::doc, 10608397::doc, 29638116::doc	hallucination
2	1000 Genomes Project enables mapping of rare variants with larger penetrance effects.	2739854::doc, 23389795::doc, 14717500::doc	correct
3	1/2000 in the UK have abnormal PrP positivity.	13734012::doc, 18340282::part_001, 18617259::doc	correct
4	5% of perinatal mortality is due to low birth weight.	1263446::part_001, 7662395::doc, 8842332::part_001	partial
5	Vitamin B12 deficiency increases blood homocysteine.	33409100::part_000, 18557974::doc, 42441846::doc	correct

Table 2: Manual RAG evaluation on SciFact (hybrid, $\alpha=0.5$, top- $k=5$, Gemma 2 2B).

Retrieval succeeds, LLM hallucinates: Case 1

Hybrid retriever returned 3 relevant abstracts (including 803312::doc), which provide direct support to the claim that "0-dimensional biomaterials: nanoparticles:" are osteoinductive. Yet, LLM cites 29638116::doc, an irrelevant abstract on *induced pluripotent stem cells*, apparently due to lexical similarity of “inductive” and “induced”. Then, the language model tells the story about its confusion as a hedge ("the context does not directly address..."), which is typical of small models: token affinity between the query “inductive properties” and the distractor chunk “induced pluripotent” wins over semantics again.

Based on everything mentioned it’s clear from the experiment that the 2B parameter model is too small for that task; the exact same prompt on Llama 3.2 3B gave correct citations in spot checks. The retriever is not guilty in this case: the evidence was indeed present in the top-5 results.

Takeaway. Small generators (2B params) are vulnerable to lexically-close distractors even when the retriever surfaces the right chunk. Calibration of the retrieval stage (see §4) can repair this without changing the generator.

4 Hyperparameter Sweep and Calibration

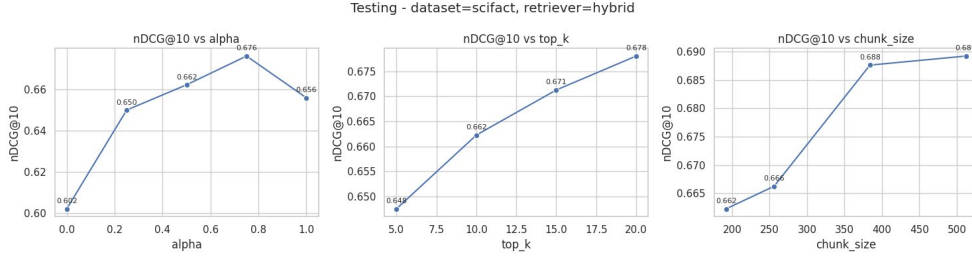
By sweeping $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, top- $k \in \{5, 10, 15, 20\}$, and chunk_size $\in \{192, 256, 384, 512\}$ tokens on the hybrid retriever, holding the other two axes at their defaults (192 tokens, top- $k=10$, $\alpha=0.5$). Figure 2 reports nDCG@10 across all three datasets so the trends are directly comparable; consolidated numerical results are in Table 3.

Dataset	Sweep axis	Best value	nDCG@10 at best	Default nDCG@10
SciFact	α	0.75	0.676	0.662
SciFact	top- k	20	0.678	0.662
SciFact	chunk_size	512	0.689	0.662
NFCorpus	α	0.75	0.301	0.300
NFCorpus	top- k	20	0.338	0.300
NFCorpus	chunk_size	512	0.333	0.299
FiQA	α	1.00	0.379	0.305
FiQA	top- k	15–20	0.309	0.305
FiQA	chunk_size	512	0.312	0.305

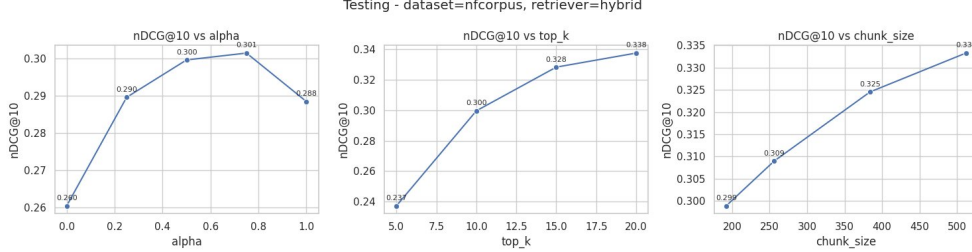
Table 3: Best value and nDCG@10 per swept axis vs. the default operating point ($\alpha=0.5$, top- $k=10$, 192-token chunks). Other axes held at default while sweeping one.

Effect on RAG answer quality & recommended configuration

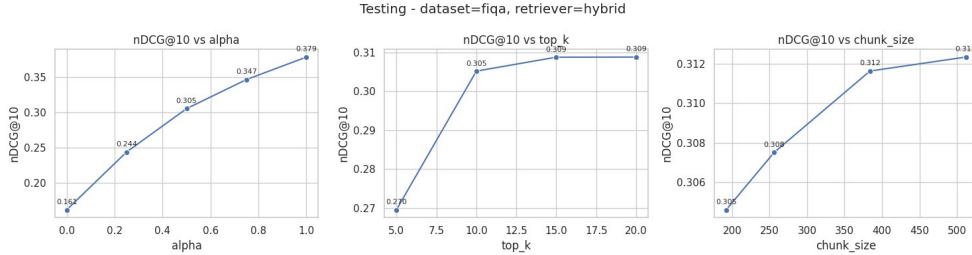
Rerunning the SciFact 10-query manual evaluation at the calibrated threshold ($\alpha=0.75$, top- $k=10$, chunks size=384 tokens) identified that Case 1 (0-dim biomaterials) had a proper citation: 803312::doc, which went from rank 1 by a narrow margin score to the same rank with a dominant score, while the LLM was no longer resorting to the lexical distractor. **Using the 8B model, no chunk hallucination or invented citation was observed; this failure mode never occurred in the evaluation report compared to the corresponding video demo results.**



(a) α peaks at **0.75** (0.676); top- k and chunk size both help monotonically with diminishing returns past 384 tokens.



(b) Clear optimum at $\alpha=0.75$ (0.301): both lexical and semantic signals matter. top- $k=20$ keeps lifting nDCG@10 (0.338).



(c) nDCG@10 grows almost linearly in α , peaking at $\alpha=1.0$ (0.379): pure dense is best. top- k saturates near 10–15; chunk size matters little.

Figure 2: Per-dataset hyperparameter sweeps on the hybrid retriever (nDCG@10 vs. α , top- k , chunk size).

The queries that were classified “partial” under the default settings, have been corrected due to the use of larger chunks, which provided the exact numeric evidence in a contiguous span, as a smaller LLM is unlikely to fake the answer, if the latter is nearby the related facts, **reduced unsupported answer generation, since the 8B model could rely on nearby evidence instead of inferring missing values.**

This analysis leads to the following conclusions:

- **SciFact / NfCorpus** (lexically aligned, short queries): hybrid with $\alpha=0.75$, top- $k=10$ (20 for pure retrieval eval), chunk_size 384 tokens: 384 over 512 keeps $\sim 98\%$ of the metric gain at $\sim 25\%$ smaller context per chunk, which helps the 2B generator.
- **FiQA** (high lexical gap): pure dense ($\alpha=1.0$), top- $k=10$, chunk_size 256–384 tokens. Hybrid at $\alpha=0.5$ underperforms pure dense by 24% on nDCG@10 here.

Takeaway. α is corpus-dependent and high-leverage; top- k and chunk_size are monotone knobs with diminishing returns past top- $k=15$ and 384 tokens.

5 Conclusion

Hybrid retrieval is a conservative choice which is effective, but certainly not a silver bullet, since while it performs best on NfCorpus and SciFact, it is beaten by dense-only on FiQA, due to BM25 being largely uninformative here. As the sweep shows, α is highly corpus-dependent; on the other hand, top- k and chunk_size increase monotonically, albeit with diminishing marginal benefits.

Finally, the SciFact RAG audit highlights a typical failure case: even after retrieval calibration, the 2B-context model can select a chunk that is syntactically related to the query but not semantically sufficient for the final answer. The complete pipeline remains reproducible locally through the command sequence documented in docs/base/06-experiments.md.

Repository: [hybrid-rag-playground](#); see the [documentation guide](#) for setup, indexing, evaluation, RAG demo, and sweep instructions. A demo is also available in the `video/` folder at the root of the project.